

Cross-Validating Binary-Time-Series-Cross-Section Data

Gokhan Ciflikli & Nils W. Metternich

CONflict Analysis LAB (COALA)

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Department of International Relations
London School of Economics and Political Science
g.ciflikli@lse.ac.uk

Department of Political Science
University College London
n.metternich@ucl.ac.uk

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Cross-validation

- ▶ Model validation technique to generalize to an independent data set
- ▶ Done by partitioning datasets into training and test sets.
- ▶ Good cross-validation strategies are important for:
 - 1 Prediction
 - 2 Preventing overfitting
 - 3 Robust inference

Binary-Time-Series-Cross-Section Data

- ▶ Very common data format (Beck et al. +2500 citations, Carter and Signorino +1000 citations)
- ▶ Binary outcome variable
- ▶ Across observations
- ▶ Over time

What is special about BTSC?

- ▶ In some instances: Special case of duration data (Beck et al. 1998)
- ▶ This is the case if underlying risk of occurrence varies over time
- ▶ Even though this means that cases are IID, risk information can only be gathered by observing complete process

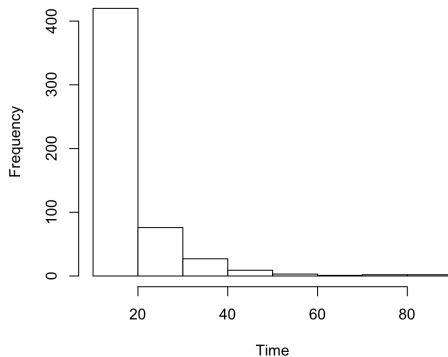
When should you care about type of CV

- ▶ If time doesn't matter you are fine
 - ▶ if it does...
- ▶ If time has a linear trend you are fine
 - ▶ if it doesn't ...
- ▶ then type of cross-validation matters

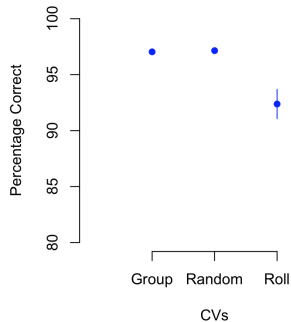
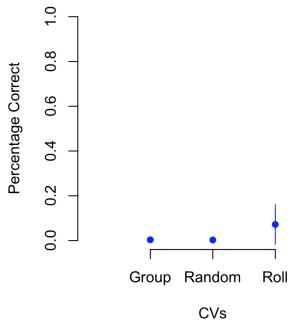
Cross-validation approaches

- ▶ Rolling-window (starting with about 60 percent of data)
- ▶ 10-fold cross-validation
- ▶ Group-wise cross-validation (10-fold)

- ▶ Survival data (Hendry, 2014)
- ▶ IDs: 1000
- ▶ Mean Life: 17
- ▶ Samples: 100



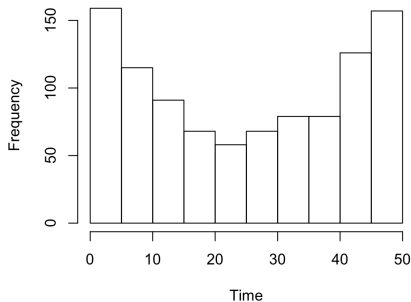
Logit: $Y=X+t+t^2+t^3$



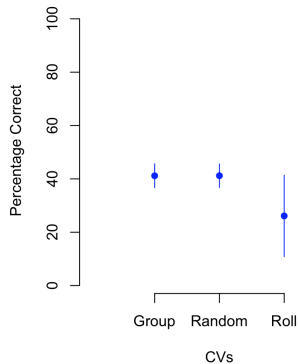
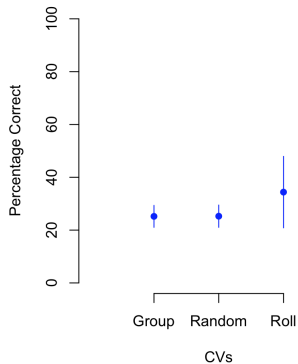
Results

Cross-validation	True-Positive Rate	True-Negative Rate
Group-wise CV (10-fold)	0.003416897	97.039709700
10-fold CV	0.002451740	97.146360996
Rolling-window CV	0.071861408	92.377115650

- ▶ Binomial with $t+t^2+t^3$
- ▶ Observations: 1000
- ▶ Samples: 100
- ▶ Similar to Carter and Signorino



Logit: $Y = X + t + t^2 + t^3$



Cross-validation	True-Positive Rate	True-Negative Rate
Group-wise CV (10-fold)	25.20300	41.17200
10-fold CV	25.26900	41.18800
Rolling-window CV	34.38667	26.12000

Other preliminary insights

- ▶ Smaller samples type of CV matters more
- ▶ Especially if risk is increasing over time
- ▶ And risk 'peaks' at particular time points

Conclusion and looking ahead

- ▶ Settle on one DGP
- ▶ Further Monte-Carlo experiments
- ▶ Currently showing differences in existing work
- ▶ Providing diagnostics for best CV type
- ▶ Understanding CV in auto-regressive data (non-IID)